RouteRL: Routing Customer-Support Tickets with Deep Q-Learning

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GitHub: `https://github.com/sidbhagat40/RouteRL`

1 Problem

Tickets routed to the wrong support team bounce between queues, wait longer and breach their Service-Level Agreements (SLAs). Production helpdesks route by the label of a text classifier (*category rule*) or by load (*round-robin*, *shortest queue*); none of these reasons jointly about classifier uncertainty, ticket urgency and team congestion. We cast routing as an MDP [7] and learn a dispatcher with a Deep Q-Network (DQN) [1], written from scratch in PyTorch.

2 Environment

A discrete-time (1 min) simulator of an 8-hour shift with C=4 categories (billing, technical, account, logistics; mix 30/35/15/20%) and K=5 teams: four specialists (3/4/2/2 agents, mean handling 6–10 min on their own category) and a Tier-1 generalist (4 agents, 10–25 min on any category). Priorities P1/P2/P3 (15/35/50%) have SLAs of 30/60/120 min. Arrivals are Poisson with a mid-shift peak (+35%); with probability 0.5 an incident doubles one category's demand for 45 min. Idle agents take breaks (1%/min, mean 15 min). An upstream classifier emits a softmax over categories from clarity$\cdot\mathbf{e}_c+\mathcal{N}(0,I)$ (top-1 accuracy ≈75%). A team that lacks the skill triages a ticket for 3 min and *bounces* it back, correctly re-tagged. Within a minute, every ticket at the dispatcher (new, held, bounced) is a separate decision, so decisions are irregular in time.

3 MDP Formulation

State $s\in\mathbb{R}^{39}$ (normalised): *ticket* – classifier probabilities (4), priority one-hot (3), SLA slack $(d_i-t)/\mathrm{SLA}_p$, re-tagged flag, #holds, #misroutes; *per team* ×5 – queue length, tickets ahead of this one by priority, idle-agent share, on-shift share (agent availability), minutes until the next agent frees; *global* – shift progress, dispatcher backlog, recent arrival rate.

Action $a\in\{0,\dots,5\}$ (discrete): route to one of the 5 teams, or *hold* the ticket for one minute.

Reward $r=-\lambda\,\Delta C$, λ=0.05, where the operating cost is

$$C=\sum_i\Big(w_{p_i}T_i+B_{p_i}\,\mathbf{1}[T_i>\mathrm{SLA}_{p_i}]+M\,m_i\Big),\quad(1)$$

T_i = time to resolution, m_i = number of misroutes, w=(3, 2, 1)/min, B=(50, 25, 10), M=5. Waiting is charged every minute, breaches when the deadline passes, misroutes at routing time.

4 Algorithm

DQN. Double DQN [2] with a dueling head [3] (MLP 39–256–256), Huber loss, Adam (3×10^{-4}), batch 256, replay 2×10^5, soft target updates (τ=0.005), ϵ-greedy $1\to0.03$ over 120k of 300k steps, demand randomised $\times[0.7,1.5]$ during training. Checkpoints are selected on the geometric mean of validation cost (5 shifts) at demand 1.0/1.3/1.45×, so a policy that only works at nominal load is never picked. Three problem-specific choices were *necessary*:

Table 1: Test results (30 shifts, ≈594 tickets each, mean ± 95% CI).

Policy	Cost	$\bar{T}$ (min)	SLA br.	P1 br.	Misr./tk
Random	126,621±5287	76.0	51.6%	21.8%	1.45
Round-Robin	115,133±5463	70.3	47.7%	20.1%	1.49
Shortest-Queue	144,620±8724	129.0	61.6%	30.6%	3.05
Category-Rule	25,961±2472	34.0	7.4%	1.3%	0.22
Skill-Aware-SQ	15,525±1068	17.0	1.3%	4.1%	**0.18**
DQN (ours)	**14,089**±812	**15.2**	**0.3%**	1.5%	0.24

(i) Semi-MDP discounting [4]: targets use $\gamma^{\Delta t}$ (γ=0.99 per elapsed minute), and every cost term *inside* a transition is discounted to its decision time (waiting $w\sum_{k<\Delta t}\gamma^k$). Discounting per decision, or charging a long wait undiscounted, makes the same wait look cheaper when split by holds – in an earlier version this drove the agent into hold loops (25–35 holds/ticket at 1.45×).

(ii) Per-ticket credit assignment. Trained on the system-wide cost stream, the DQN cannot tell which decision caused which cost and degenerates into holding loops (episodes grow from ∼700 to >10,000 decisions). Every term of C belongs to one ticket, so $C=\sum_i C_i$ exactly. We therefore let each transition follow one ticket from a decision to its next decision (after a hold/bounce) or to resolution: $y=-\lambda C_i^{[t,t')}+\gamma^{t'-t}\,Q_{\bar\theta}(s',\arg\max_{a'}Q_\theta(s',a'))$, with one shared network for all tickets (as in multi-driver order dispatching [5]). Because purely "selfish" per-ticket costs ignore congestion externalities [6], we add a marginal-cost (Pigouvian) term during training only: a ticket occupying one of n_k agents for S minutes is charged $\frac{S}{n_k}\sum_{j\in\mathrm{queue}_k}w_{p_j}$.

(iii) Hold cap with action masking. A ticket may be held at most 5 times; the illegal hold is masked both when acting and in the target's arg max (masks are stored in replay).

Baselines. Random; Round-Robin and Shortest-Queue (category-blind, as proposed); Category-Rule (send to the classifier's top-1 specialist); Skill-Aware-SQ (least-loaded team per available agent among teams able to handle the predicted category) – a strong hand-designed heuristic. All policies are evaluated on 30 held-out shifts with *identical* ticket streams.

5 Results

Headline (Table 1, Fig. 1). The DQN cuts operating cost by 88% vs. Round-Robin, 90% vs. Shortest-Queue, 46% vs. Category-Rule and 9.3% vs. Skill-Aware-SQ, which it beats on 29/30 paired shifts (paired saving 1436 ± 349, t=8.1). SLA breaches fall 4× (1.3%→0.3%), P1 breaches 2.7× (4.1%→1.5%), mean resolution time from 17.0 to 15.2 min. Across three training seeds the cost is 14,131 ± 76 (std).

What was learned (Fig. 2). The agent discovered priority-aware use of the slow generalist: it sends only 6% of P1 tickets there (heuristic: 18%) but 23% of P3 tickets (20%), reserving fast specialists for urgent work – a $c\mu$-like rule that was never hand-coded. Its misroute rate is higher (0.24 vs 0.18 per ticket), but the extra penalties are outweighed by shorter waits and fewer breaches.

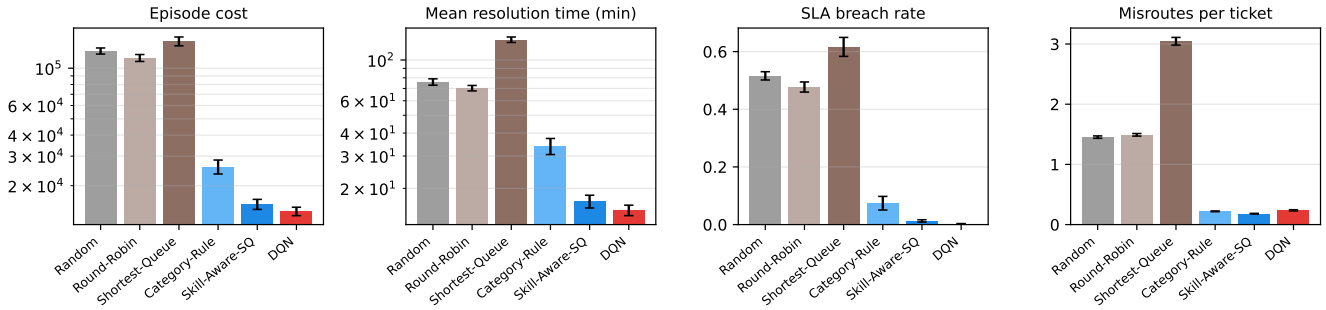


Figure 1: Test metrics per policy (30 shifts, nominal demand, 95% CI). Cost and resolution time are on a log scale.

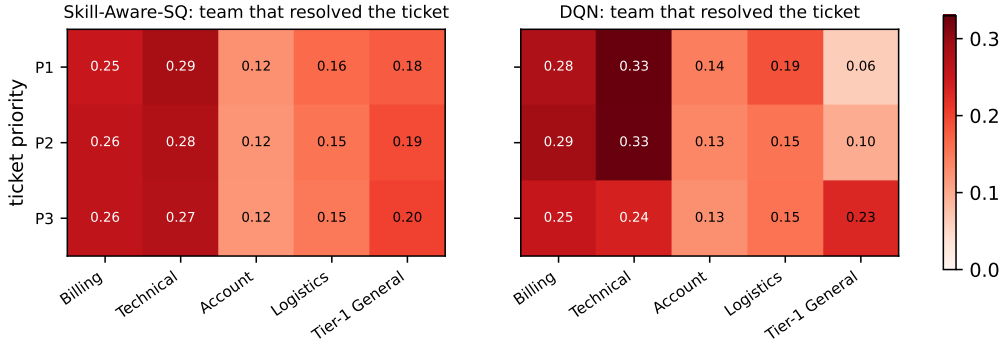


Figure 2: Share of each priority's tickets resolved by each team: the DQN keeps P1 tickets on the fast specialists and sends low-priority overflow to the slow Tier-1 generalist.

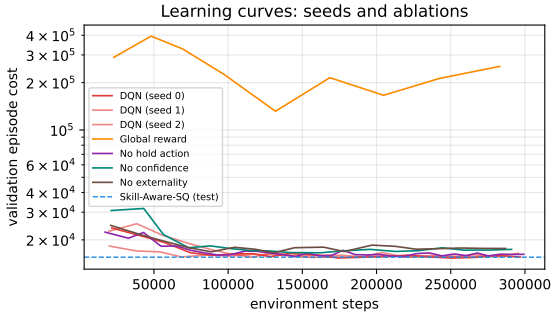


Figure 3: Validation cost during training: three seeds and ablations. The global-reward DQN never learns; dashed line: Skill-Aware-SQ.

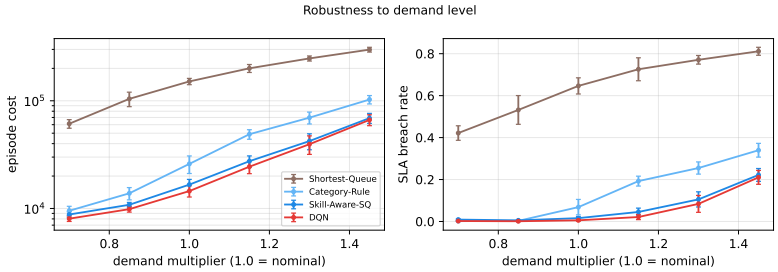


Figure 4: Cost and SLA-breach rate vs. demand (10 shifts per point). The DQN is the cheapest policy at every demand level.

Ablations (Fig. 3; nominal / 1.45× demand, 30 test shifts). *Global reward* (same DQN on the system-wide cost stream): cost 126,158, 56% breaches – credit assignment is essential. *No confidence* (top-1 label only): +3.4% cost. *No externality*: +3.1% at nominal and +9.3% at 1.45× (25% vs 19% breaches), so the Pigouvian term matters under congestion. *No hold*: −2.5% / −4.0%, i.e. with the fixes above the hold action is not needed; we keep it in the main agent for comparability.

Robustness (Fig. 4). The DQN is the cheapest policy at every demand level from 0.7× (−8.5% vs. Skill-Aware-SQ) to 1.45× (−3.7%). On the 30 test shifts at 1.45× all three seeds beat Skill-Aware-SQ (60.8k–62.6k vs. 64.2k) with at most 1.8 holds/ticket; before fixes (i) and (iii) and multi-load selection the agent was 18% worse there.

6 Discussion & Limitations

RL pays off where heuristics are blind: classifier confidence, urgency and heterogeneous service rates. Its margin shrinks under extreme overload, where per-ticket credit assignment remains partly selfish; better externality estimates or centralised critics are natural next steps. The simulator's parameters are synthetic; with real logs they would be fit from ticket histories.

References

[1] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, 518:529–533, 2015.

[2] H. van Hasselt, A. Guez, D. Silver, "Deep reinforcement learning with double Q-learning," *AAAI*, 2016.

[3] Z. Wang et al., "Dueling network architectures for deep reinforcement learning," *ICML*, 2016.

[4] S. J. Bradtke, M. O. Duff, "Reinforcement learning methods for continuous-time Markov decision problems," *NIPS*, 1995.

[5] X. Tang et al., "A deep value-network based approach for multi-driver order dispatching," *KDD*, 2019.

[6] T. Roughgarden, É. Tardos, "How bad is selfish routing?," *J. ACM*, 49(2):236–259, 2002.

[7] R. S. Sutton, A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.